EXP:4	Virtual Private Server (VPS) Provisioning and Configuration
5/8/2026	

AIM

To provision a Virtual Private Server (VPS) using VirtualBox, install Ubuntu, configure network settings (IP Address, Subnet Mask, Gateway, DNS), and enable SSH remote access.

OBJECTIVES

- To understand the concept of Virtual Private Server (VPS).
- To create and manage virtual machines using VirtualBox.
- To install Ubuntu Server OS.
- To configure static network settings.
- To enable SSH remote access.

SOFTWARE REQUIREMENTS

- Oracle VirtualBox
- Ubuntu Server 22.04 LTS ISO (as specified in the lab manual)
- Windows/Linux Host System
- Internet Connection
- Terminal Access

THEORY

A Virtual Private Server (VPS) is a software-based server created using virtualization technology. It operates independently with its own operating system, storage, memory, and network configuration while sharing the physical hardware with other virtual machines. Virtualization enables efficient resource utilization and forms the foundation of cloud computing platforms.Network configuration parameters:

Parameter	Description
IP Address	A unique address assigned to a device on a network.
Subnet Mask	Determines the network and host portions of an IP address.
Gateway	Provides access to networks outside the local network.
DNS	Converts domain names into IP addresses.
SSH	A secure protocol used for remote administration of Linux systems.

PROCEDURE

Step 1: Create Virtual Machine

Parameter	Value
Name	VPS-Server
OS Type	Ubuntu (64-bit)
RAM	2 GB
CPU	2 Cores
Storage	20 GB

Attach the Ubuntu ISO, start the VM, follow the installation wizard, create a username and password, and complete the installation.

Step 2: Update Ubuntu

```
sudo apt update
sudo apt upgrade -y
```

Step 3: Identify Network Interface

```
ip a
```

The practical execution identified the active interface as enp0s3 with the initial NAT address 10.0.2.15/24.

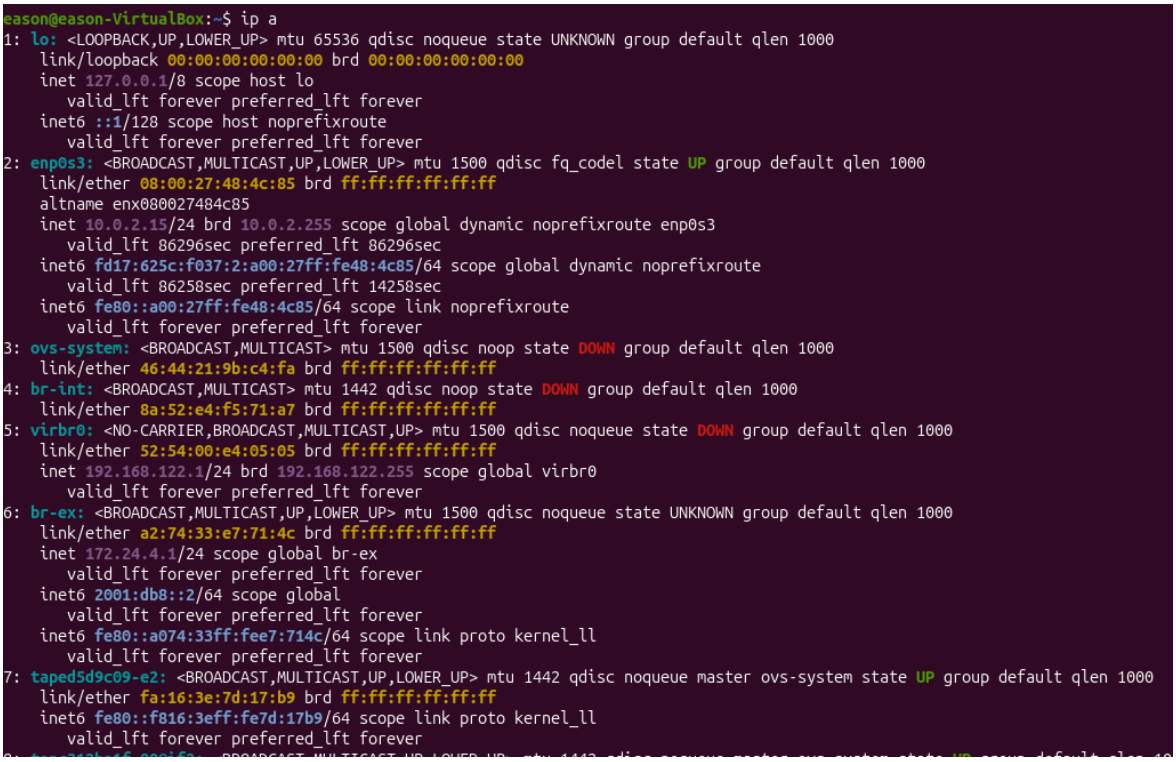


Figure 1. Network interface identification using ip a.

Step 4: Check Routing Information

```
ip route
```

The VirtualBox NAT gateway was identified as 10.0.2.2 through enp0s3.

```
eason@eason-VirtualBox:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
172.24.4.0/24 dev br-ex proto kernel scope link src 172.24.4.1
192.168.122.0/24 dev virbr0 proto kernel scope link src 192.168.122.1 linkdown
```

Figure 2. Routing table showing the VirtualBox NAT gateway.

Step 5: Configure Static IP

```
sudo nano /etc/netplan/00-installer-config.yaml
```

The following configuration was used for the actual VirtualBox NAT network:

```
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 10.0.2.100/24
      routes:
        - to: default
          via: 10.0.2.2
      nameservers:
        addresses:
          - 8.8.8.8
          - 8.8.4.4
```

```
eason@eason-VirtualBox:~$ sudo cat /etc/netplan/00-installer-config.yaml
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 10.0.2.100/24
      routes:
        - to: default
          via: 10.0.2.2
      nameservers:
        addresses:
          - 8.8.8.8
          - 8.8.4.4
```

Figure 4. Saved Netplan configuration verified with sudo cat.

Step 6: Apply Network Configuration

```
sudo netplan apply
```

The command displayed a systemd-networkd reload warning because the desktop environment was not running systemd-networkd in the expected way. Subsequent routing and connectivity checks confirmed that the static configuration was nevertheless active.

Step 7: Verify Network Configuration

```
ip addr
ip route
cat /etc/resolv.conf
```

The verified practical values were: interface enp0s3, static IP 10.0.2.100/24, default gateway 10.0.2.2, and DNS servers 8.8.8.8 and 8.8.4.4.

Step 8: Test Connectivity

```
ping -c 4 10.0.2.2
ping -c 4 8.8.8.8
ping -c 4 google.com
```

The gateway, Internet address, and DNS name were all reachable. The practical screenshots showed 4 packets transmitted and 4 received with 0% packet loss for the gateway and 8.8.8.8, and successful DNS resolution of google.com followed by successful replies.

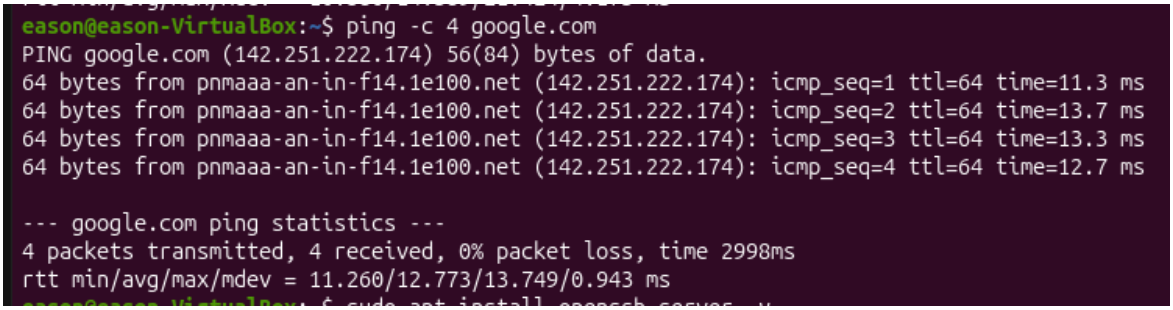


Figure 6. Internet and DNS connectivity verification.

Step 9: Install OpenSSH Server

```
sudo apt install openssh-server -y
```

Step 10: Enable and Start SSH Service

```
sudo systemctl enable ssh
sudo systemctl start ssh
sudo systemctl status ssh
```

The SSH service was verified as active (running), enabled, and listening on port 22.

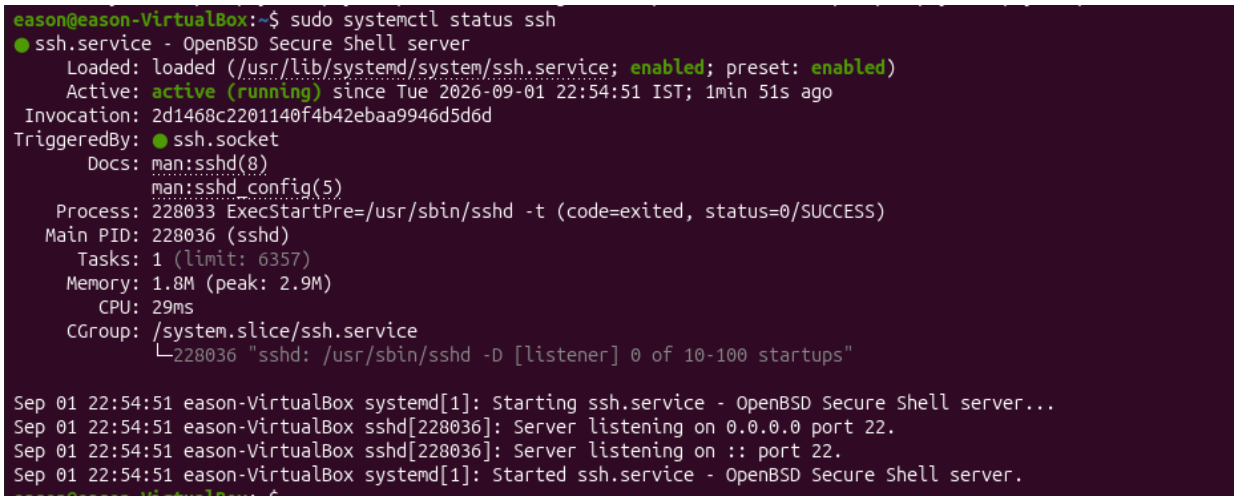


Figure 7. OpenSSH service status showing active (running) and port 22 listening.

Step 11: Remote SSH Login

```
ssh eason@10.0.2.100
```

The SSH connection was successfully established to the static IP address. The Ubuntu welcome message appeared after authentication, demonstrating successful remote login.

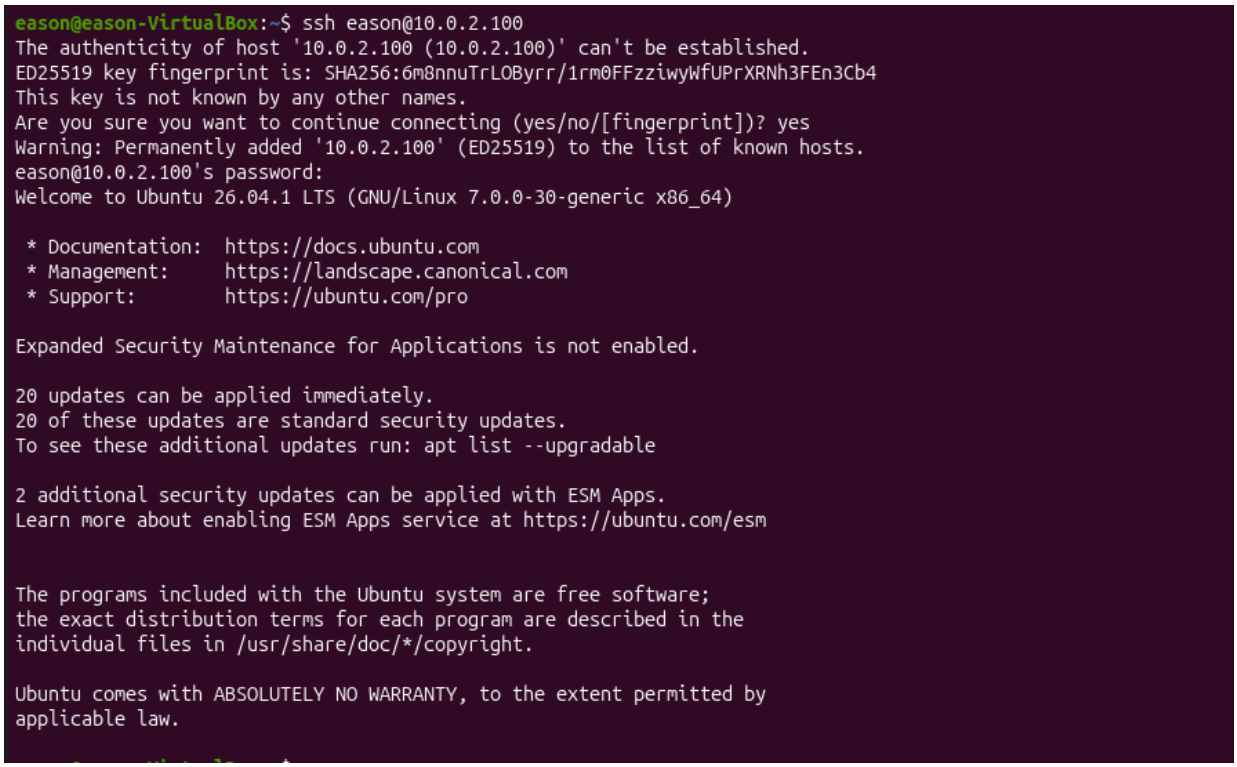


Figure 8. Successful SSH remote login to 10.0.2.100.

COMMANDS USED

```
sudo apt update
sudo apt upgrade -y
ip a
ip route
sudo nano /etc/netplan/00-installer-config.yaml
sudo cat /etc/netplan/00-installer-config.yaml
sudo netplan apply
ip addr
cat /etc/resolv.conf
ping -c 4 10.0.2.2
ping -c 4 8.8.8.8
ping -c 4 google.com
```

```
sudo apt install openssh-server -y
sudo systemctl enable ssh
sudo systemctl start ssh
sudo systemctl status ssh
ssh eason@10.0.2.100
```

OBSERVATIONS

Parameter	Status
VM Creation	Successful
Ubuntu Installation	Successful
Network Interface	enp0s3 identified
Static IP Configuration	Successful — 10.0.2.100/24
Subnet Mask	Successful — /24 (255.255.255.0)
Gateway Configuration	Successful — 10.0.2.2
DNS Configuration	Successful — 8.8.8.8, 8.8.4.4
Gateway Connectivity	Verified — 0% packet loss
Internet Connectivity	Verified — 0% packet loss to 8.8.8.8
DNS Resolution	Verified — google.com resolved successfully
SSH Installation	Successful
SSH Service	Verified — active (running), port 22
Remote Login	Successful — ssh eason@10.0.2.100

EXPECTED OUTPUT

- VPS virtual machine created successfully in VirtualBox.
- Ubuntu environment installed and running.
- Static IP, subnet mask, gateway, and DNS configured.
- Internet and DNS connectivity verified.
- OpenSSH Server installed and running.
- SSH remote login completed successfully.

CONCLUSION

A Virtual Private Server (VPS) environment was successfully created using Oracle VirtualBox. The Ubuntu VM was configured with a static IP address (10.0.2.100/24), gateway (10.0.2.2), and DNS servers (8.8.8.8 and 8.8.4.4). Network connectivity was verified, OpenSSH was installed and enabled, and secure remote access was successfully demonstrated using SSH.